Lecture - 18

* Incorrect view of large signal and small signal models:

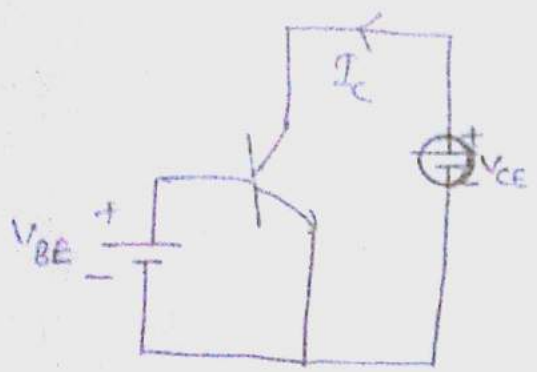
Large signal model — DC

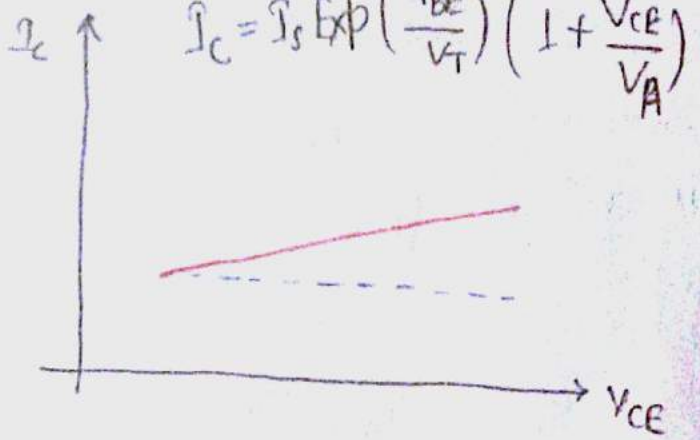
Small signal model — AC

→ The large signal model is a general model that allows us to analyse the ckt when we have not necessarily small swings/small change. (not much less than $26mV = V_T$).

→ The small signal model assumes small perturbations in the bias conditions.

* Early Effect:-





$$I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right)\left(1 + \frac{V_{CE}}{V_A}\right)$$

Official method of small signal model derivation:-

Step I:- Assume the transistor is biased at a ~~center~~ certain operating point.

Step II:- Apply a voltage change between two terminals and measure the resulting current changes

Step III:- Model the current changes by proper electric devices.

g_m: $g_m = \frac{dI_C}{dV_{BE}} = \left[\frac{I_S}{V_T}\exp\left(\frac{V_{BE}}{V_T}\right)\right]\left(1 + \frac{V_{CE}}{V_A}\right) = \frac{I_C}{V_T}$

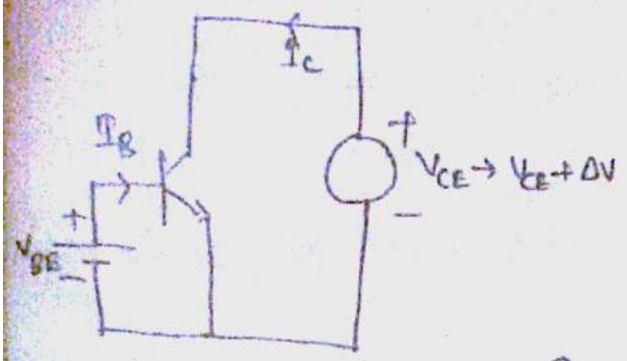
r_π: $I_B = \frac{I_C}{\beta} \Rightarrow \Delta I_B = \frac{\Delta I_C}{\beta} \Rightarrow r_\pi = \beta / g_m$

g_m and r_π have not changed.

Scanned by CamScanner

* Previous small signal equivalent model is not correct for this equation of collector current.

* Inclusion of Early Effect:-



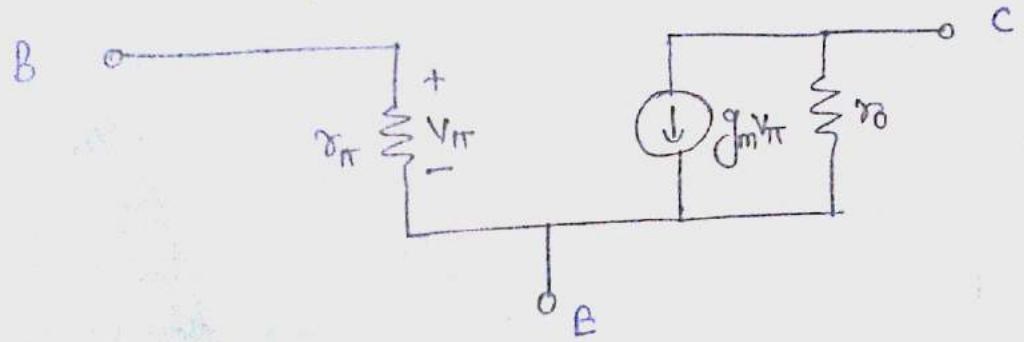
* Let's suppose the collector emitter voltage V_{CE} makes this change and goes to $V_{CE}+\Delta V$.

So now I_C will be

$$I_C + \Delta I_C = \underbrace{I_S \exp\left(\frac{V_{BE}}{V_T}\right)\left(1+\frac{V_{CE}}{V_A}\right)}_{I_C} + \underbrace{I_S \exp\left(\frac{V_{BE}}{V_T}\right)\left(\frac{\Delta V}{V_A}\right)}_{\Delta I_C}$$

$$\Rightarrow \Delta I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right)\frac{\Delta V}{V_A}$$

→ ΔI_C is linearly proportional to ΔV as V_{BE} is constant. So for a given base emitter voltage V_{BE}, if we increament the V_{CE} by ΔV then collector current increases by ΔV.



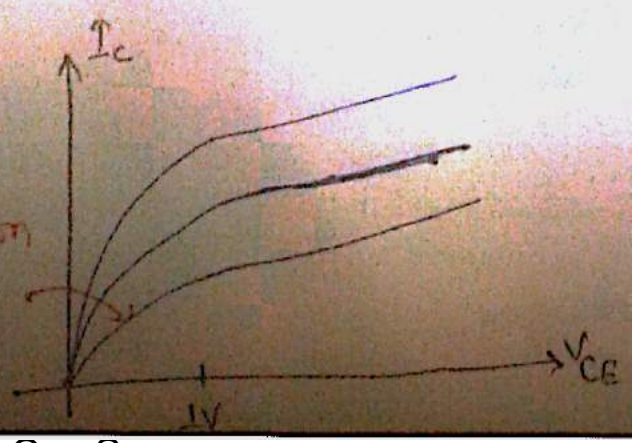
The complete small signal model in effect of Early effect

Since ΔI_C is linearly dependent on ΔV so for this we model a resistor namely r_o as

Output resistance →

$$\boxed{r_o = \frac{V_A}{I_S \exp\left(\frac{V_{BE}}{V_T}\right)} \simeq \frac{V_A}{I_C}}$$

Note:-



When we let V_{CE} collapses then what happens?

Scanned by CamScanner

* Let say $V_{BE} = 800mV$ and V_{CE} starts at 1V. Since BE junction is FB and collector Base junction is RB. So collector point is 200mV more positive than base. so the device is in Forward Active region.

* If we let V_{CE} comes down to 800mV then V_{CE} & V_{BE} both at 800mV so that's ok. But If V_{CE} comes more down the Base collector becomes FB. The det does not remains in forward active region. Therefore the current I_C drops. So the circuit is in saturation which is need to be avoided.

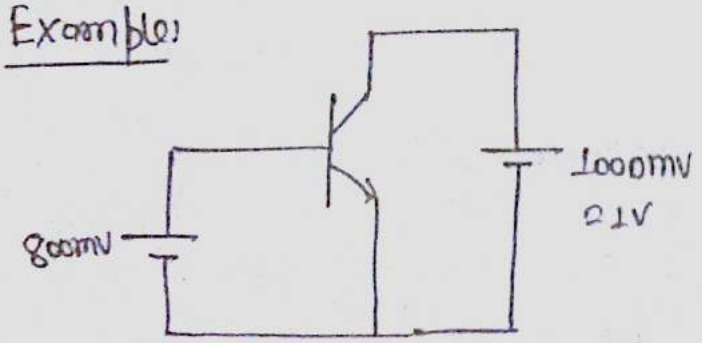
* When V_{CE} comes down to 800mV then it goes into saturation, to avoid saturation.

$V_{CE} \geq 800mV$ ← for avoiding Saturation

* Soft saturation: It is acceptable,

$$500mV < V_{CE} < 800mV$$

Example:



It is in forward active region.

$$I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right)$$

$$I_C = 5\times10^{-16} \exp\left(\frac{800}{26}\right)$$

$$= 11.53\ mA$$

$I_S = 5\times10^{-16}\ A$,

$\beta = 100$, $V_A = 5V$

$$g_m = \frac{I_C}{V_T} = 0.445 \mho = \frac{1}{2.25\Omega}$$

$$r_\pi = \frac{\beta}{g_m} = 227.273\Omega$$

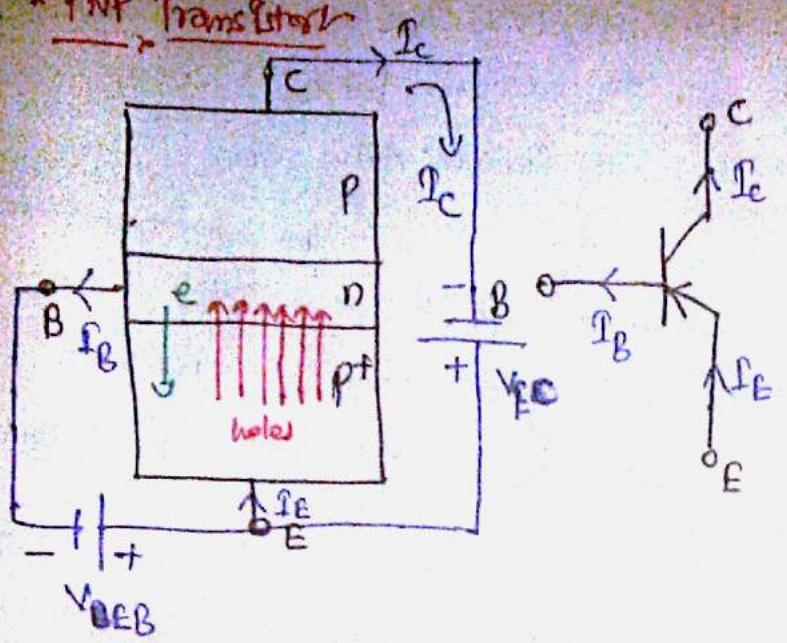
$$r_o \approx \frac{V_A}{I_C} = \frac{5}{11.53\ mA} = 433.65\Omega$$

Note: $g_m r_o >> 1$

from these values we can model small signal model.

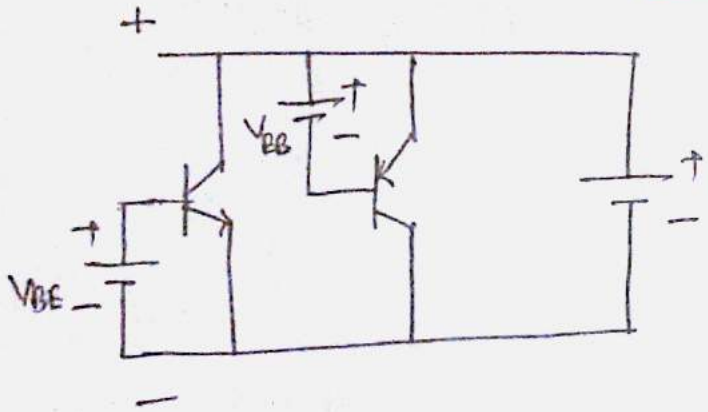
Scanned by CamScanner

* PNP Transistor



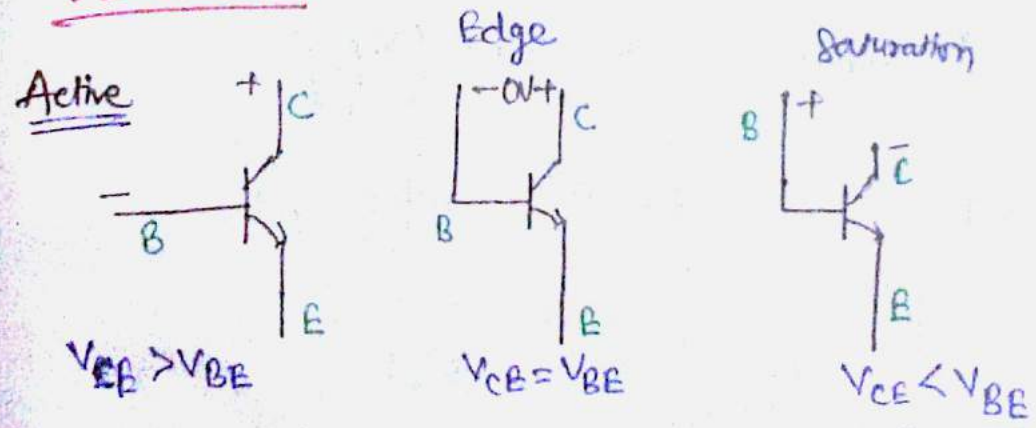
* In this just the polarity of voltages are reverse of npn transistor.
* All the operations are same as npn transistor.

* Forward Active region: $V_{EB} > 0$, $V_{EC} \leq V_{EB}$

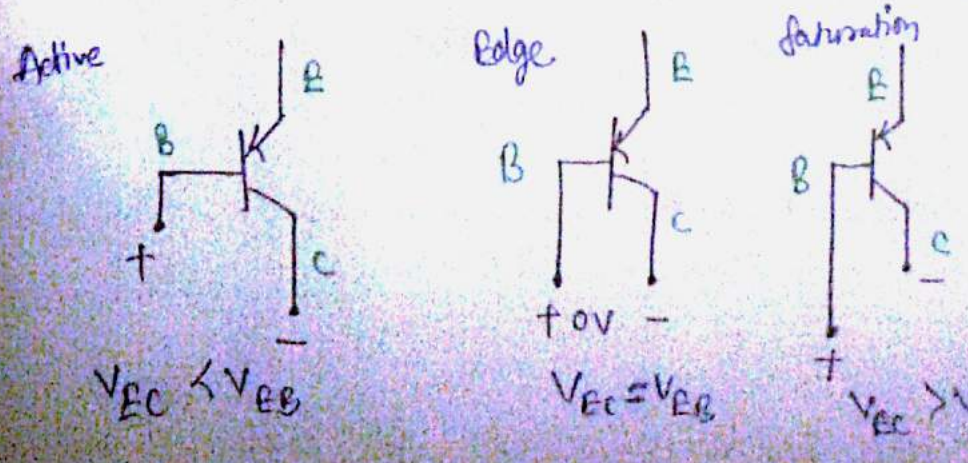


← Visualisation how npn & pnp are connected.

Visualization:-



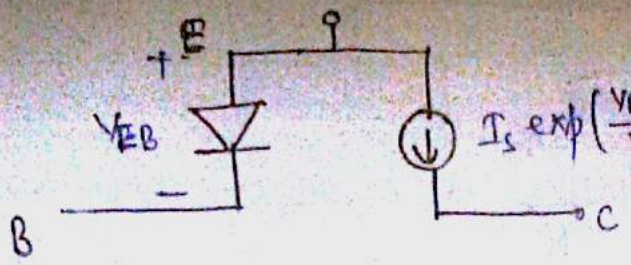
NPN



PNP

Scanned by CamScanner

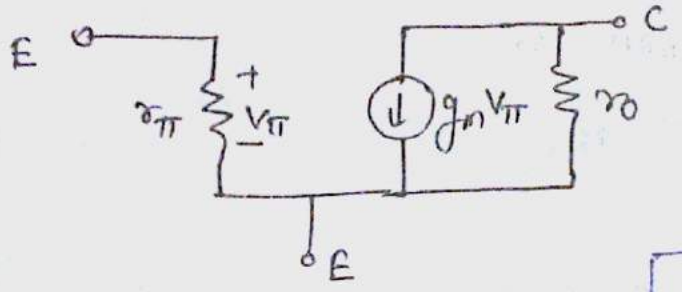
Large signal model of PNP transistor:-



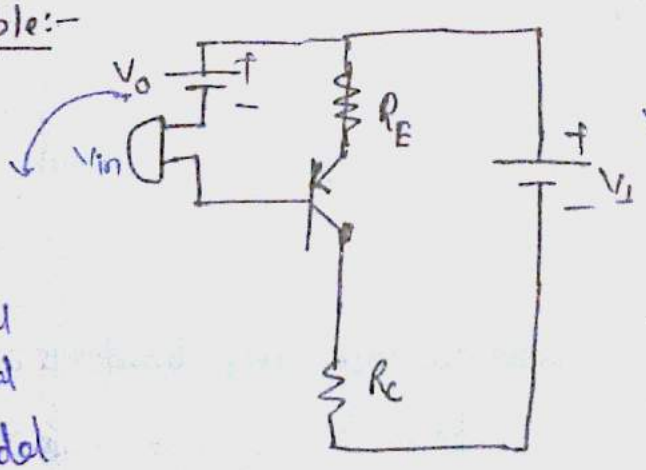
$\left(1+\frac{V_{EC}}{V_A}\right)$ where takes its early effect

Lecture – 19

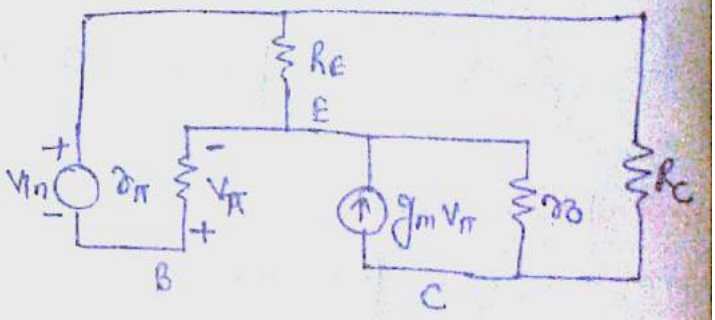
Small signal model of PNP transistor:- The small signal model of PNP transistor is identical to that of NPN transistor.



Example:-

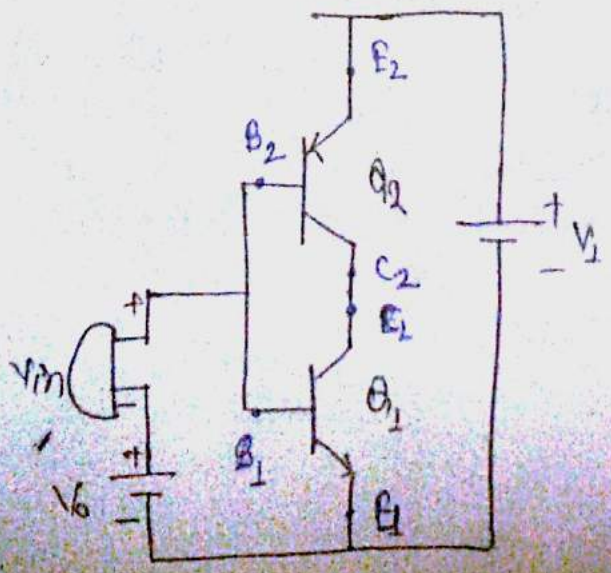


S.C. for small signal model



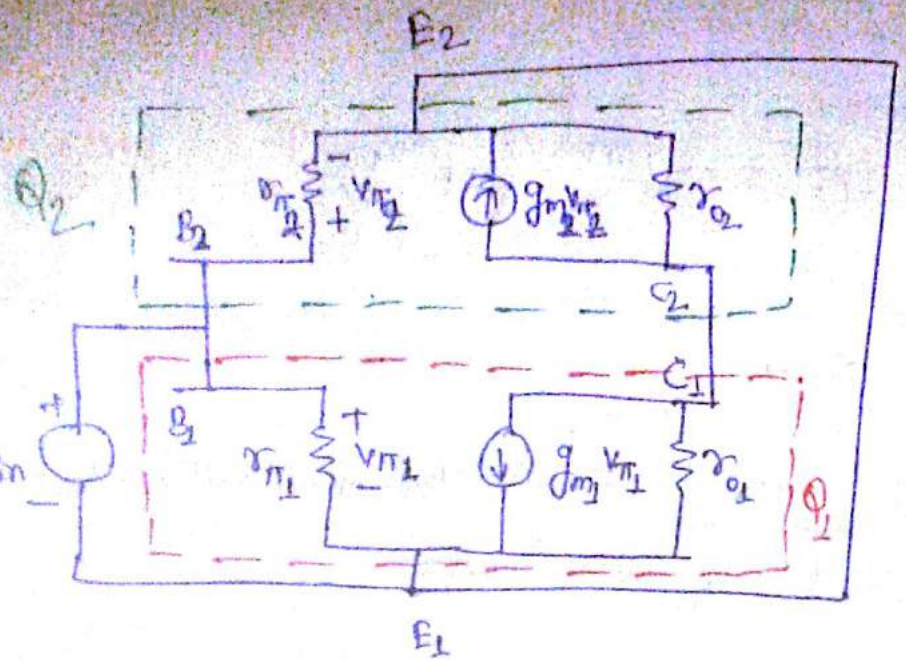
Small signal model

Quiz



Draw small signal model

Scanned by CamScanner



→ r_{o_1} is in parallel with r_{o_2}.

→ r_{π_1} is in parallel with r_{π_2}.

→ $V_{\pi_1} = V_{\pi_2}$

→ $g_{m_1} V_{\pi_2}$ is in parallel with $g_{m_2} V_{\pi_2}$.

∴ Amplifiers —

— Design procedure

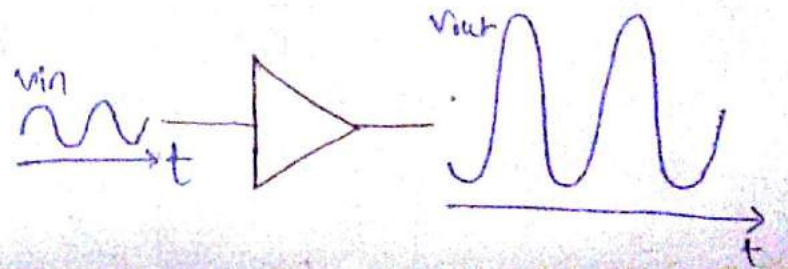
(a) Select a topology

(b) Create proper bias conditions for all transistors.

(c) Analyse the small signal properties of the amplifier and see if they are satisfactory.

— Amplifier characteristics

Amplifier takes an signal and generates a output signal same as input signal but in greater magnitude.



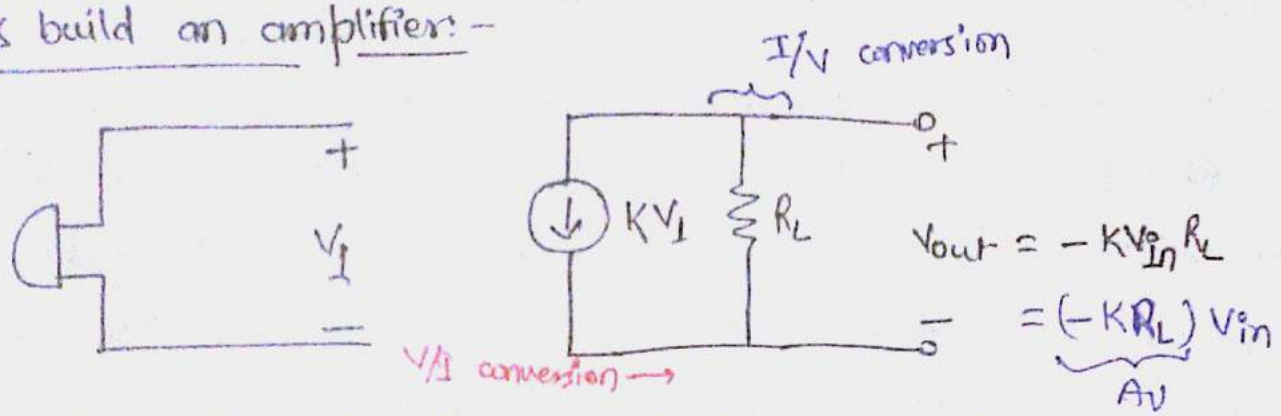
Scanned by CamScanner

Voltage Gain:- The voltage gain is

$$\text{Voltage gain} = \frac{\text{Output swing}}{\text{Input swing}} = A_v$$

Power Consumption:- When we have an amplifier it has transistor in this, transistor is biased even with no signal it needs a current. That current is coming from the battery. So bias current drain the battery.

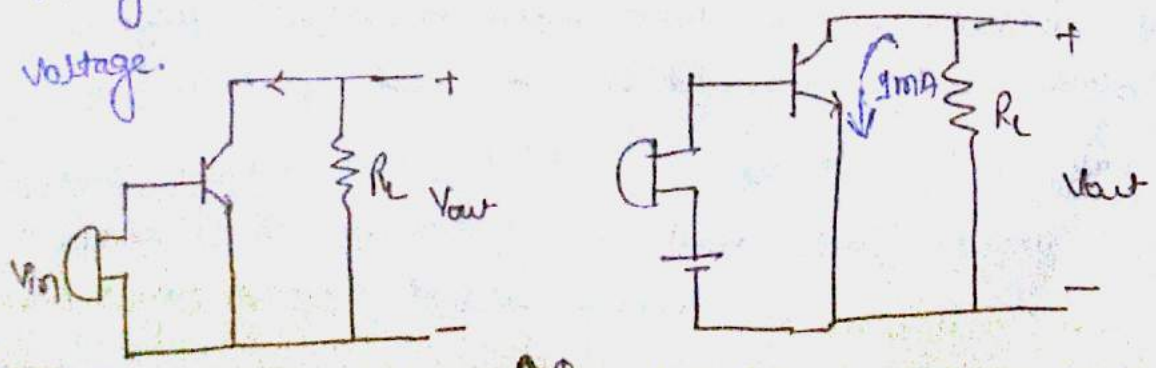
Lets build an amplifier:-

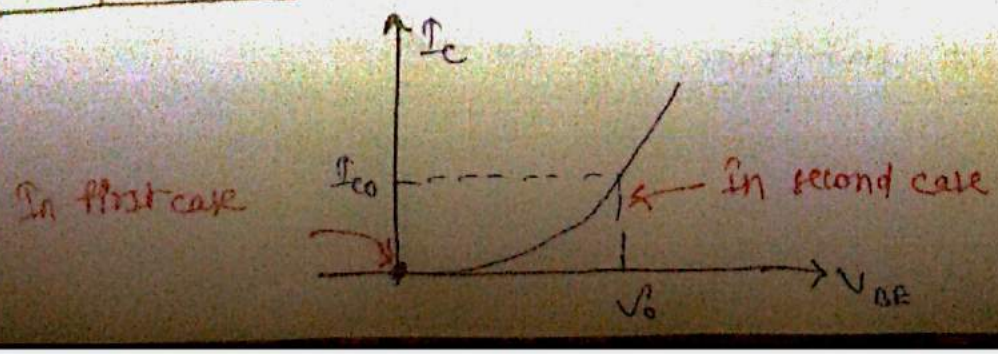


$$V_{out} = -KV_{in}R_L$$
$$= \underbrace{(-KR_L)}_{A_v} V_{in}$$

→ In this gain is −ve that means if we have a sine wave and amplified it then we just get an amplified & flipped version of sine wave.

→ In this, voltage dependent current source converts the voltage into current and the resistor converts the current into voltage.





Scanned by CamScanner

→ Circuit in first case does not work, because there is no bias current in the device. If we just connect microphone directly to the base emitter and microphone voltage changes by very small amount (5-10 mV) and the current I_c is very very small. So it is not usable.

→ In the second ckt, microphone is connected in series with a battery V_0. So Even if the microphone voltage is 0 then we have some bias current. Once we have certain bias current then we have certain g_m. In the first case, g_m is zero. The transconductance tells us that how well the transistor converts a voltage to the current.

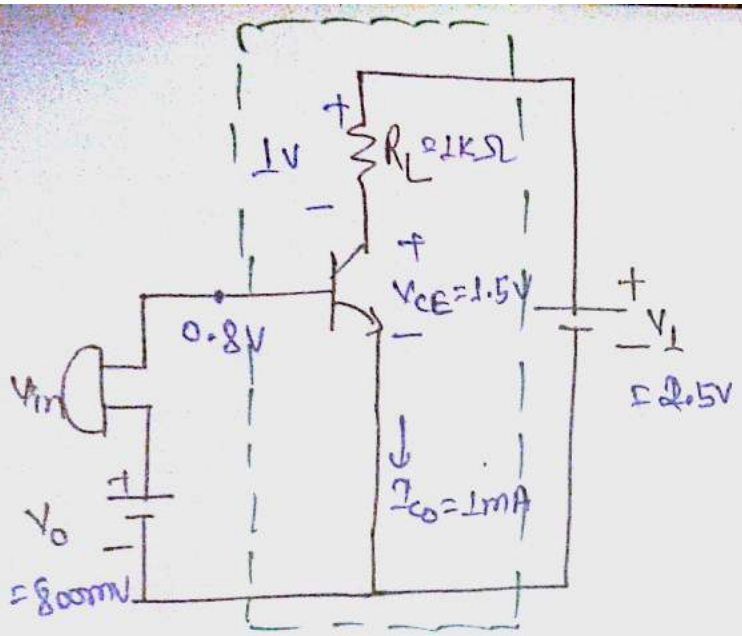
→ In the second case, this ckt is not acceptable for amplifier. Since for the amplifier, the transistor needs to be in forward active region. In this ckt, Base-emitter is forward biased but collector base must not be forward biased. In this topology, we have certain amount of current (say 1mA). But it does not have the path to flow through. Now let's say we have a current like that then

$$V_{out} = (-\overset{1mA}{I_{co}})\overset{1K\Omega}{R_L}$$

$$V_{out} = -1V$$

That means collector voltage with respect to emitter voltage is -1V. Since collector voltage should be at least equal to base voltage. If base-E Voltage V_{BE} is 800mV then collector should be 800mV greater than emitter. So the device is dead. It does not have +ve & -ve terminal so that current flow through because current flow from +ve to -ve.

Scanned by CamScanner



Common – Emitter Topology

* The resistor is connected here for I/V conversion.

By letting the values of parameters,

$V_{CE} + 1V = 2.5V = V_1$

$V_{CE} = 1.5V$ by KVL

$V_{CE} > V_{BE}$

So CB junction is RB without the battery V_1 it will not be RB.

So V_0 and V_1 both are necessary for biasing.

→ This is called as Common Emitter Topology because emitter is common to both input and output.

Scanned by CamScanner